Kaustubh Sakhalkar

✉ sakhalkark@gmail.com ✉ s8kasakh@stud.uni-saarland.de 💻 sakhalkark.github.io
https://github.com/sakhalkark @cos_cube_ LinkedIn Google Scholar

Research Interests

I am interested in computer vision, machine learning, optimization and image synthesis. Specifically, my research interests are in unsupervised and self-supervised learning, video understanding and deep learning.

Education

2019 – Present	Saarland University, Saarbrücken, Germany
	MSc in Visual Computing
2014 – 2018	Vellore Institute of Technology
	B.Tech (Computer Science with a specialization in bioinformatics)
	CGPA: 8.26/10
2014	Higher Secondary Certificate (Maharashtra State Board) 79.23%
2012	Secondary School Certificate (Maharashtra State Board) 91.82%

Work Experience

06/2018 – 06/2019	DreamVu Inc.
	Applied Computer Vision Engineer.
01/2018 – 05/2018	INRIA Sophia-Antipolis
	Research Intern - STARS team
	Supervisor - Francois Bremond
05/2017 – 07/2017	Indian Statistical Institute – Kolkata
	Research Intern - CVPR Unit
	Supervisor - Ujjwal Bhattacharya
12/2016 – 01/2017	Gesture Research
	Computer Vision Intern

Publications

"*A new hybrid architecture for human activity recognition from rgb-d videos.*" Srijan Das, Monique Thonnat, Kaustubh Sakhalkar, Michal Koperski, Francois Bremond, and Gianpiero Francesca. In International Conference on Multimedia Modeling, pp. 493-505. Springer, Cham, 2019 (MMM). [pdf]

"*Spatio-Temporal Grids for Daily Living Action Recognition.*" Srijan Das, Kaustubh Sakhalkar, Michal Koperski, and Francois Bremond. In Proceedings of the 11th Indian Conference on Computer Vision, Graphics and Image Processing, pp. 1-6. 2018 (ICVGIP). [pdf]

"Learning to Represent Spatio-Temporal Features for Fine Grained Action Recognition." Kaustubh Sakhalkar, and François Brémond. In 2018 IEEE International Conference on Image Processing, Applications and Systems (IPAS), pp. 268-272. IEEE, 2018. [pdf]

Technical Skills

Languages : Python, C++.

Libraries : SQL, OpenCV, Tensorflow, Pytorch, Keras, Numpy, Scikit-Learn, Pandas, CUDA, Caffe.

Others : LaTeX, Bash, Linux.

Academic Achievements

- Selected for Machine Learning Summer School at Tübingen Germany (MLSS2020). (180 out of 1200 applicants selected) [link]
- Chancellors Special achievers award for exemplary performance at national / international events and meritorious academic performance awarded to students for all round excellence. (VIT)
- Selected for vision understanding and machine intelligence summer school at INESCTEC Portugal. (2017) (did not attend)
- Second position in WeHack hackathon at VIT University.
- Ingenius National hackathon finalist (top 10 out of 300 teams)
- General Electric healthcare hackathon finalist

Other Positions of Responsibility

2016-2017	Hostel Review Committee Member (VIT University) Student representative for hostel block.
2016-2018	Student member of Creations Labs and SEDS-VIT (Students for the Exploration and Development of Space)

Relevant Courses

VIT University: Object Oriented Programming, Computer Architecture and Organization, Data Structures and Algorithms, Operating Systems, Soft Computing, Agent Based Intelligent Systems, Database Concepts, Image and Vision Computing, Probability and Statistics, Software Engineering, Data Warehousing and Data Mining, Linear Algebra, Automata Theory, Computer Networks.

Saarland University: Neural Networks, Machine Learning, Image Acquisition Methods, Realistic Image Synthesis, Continuous Optimization, Image Processing and Computer Vision, Seminar on Computer Vision and Machine Learning for Computer Graphics, High Level Computer Vision.

Projects

DreamVu Inc.

Developed a depth and disparity estimation pipeline for stereo images in omnidirectional cameras.
Wrote application software for crowd monitoring and people detection on Jetson TX2.

Multimodal data action recognition (INRIA)

Studying the contribution of using multiple video cues for representing appearance, short-term and pose based motion to reach a high action recognition rate.
Proposed multimodal data feature fusion strategy for disambiguating similar actions with sequential refinement.

Spatio-temporal action grids (INRIA)

Addressing the challenge of spatio-temporal relationship in videos for action recognition. Introducing grids in the spatial domain as well as temporal domain for spatio-temporal modelling of features to represent temporal information, by dividing the action sequence into temporal segments termed as temporal grids. Cross subject variance is targeted by applying clustering on the appearance features and velocity features to partition the temporal segments. Results were competitive with the state of the art for NTU-60 dataset.

Speeded up 3D convolutions for action recognition (INRIA)

Designed a factorized 3D convolutional block for efficient action recognition achieving faster training time. Improved optical flow representation by using gradients of magnitude and orientation. Achieving improved action recognition accuracy compared to baseline model and faster runtime.

Document image segmentation (Indian Statistical Institute - Kolkata)

Proposed a novel method for degraded document image character segmentation using a convolutional autoencoder with skip connections and ensemble of three global and local thresholding methods. Learned weighted voting for the decisions from three pipelines to obtain a binary decision for each pixel. Patch to patch binarization technique for document page. First auto-encoder based method for document segmentation achieving results close to the state of the art.

European Rover Challenge (SEDS, VIT University)

Developed a stereo camera visual odometry system for the rover using LK optical flow and FAST algorithm also developed the control interface using Qt GUI tools. Our team was finally placed 20th globally and 2nd in Asia.

Expression recognition in Videos (Saarland University)

Designed a network for body expression recognition from video and human pose on BOLD dataset by learning correlations between emotions.
Two stream network for fusing pose and video features.

Miscellaneous Projects

- Text classification using recurrent neural networks with attention mechanism.
- Low cost device for non-invasive oral cancer detection for rural population using colour segmentation.
- Image deblurring using proximal gradient descent.
- Daily living activity recognition dataset curation.
- Development of image quality assessment tools.

Transcripts available on request.